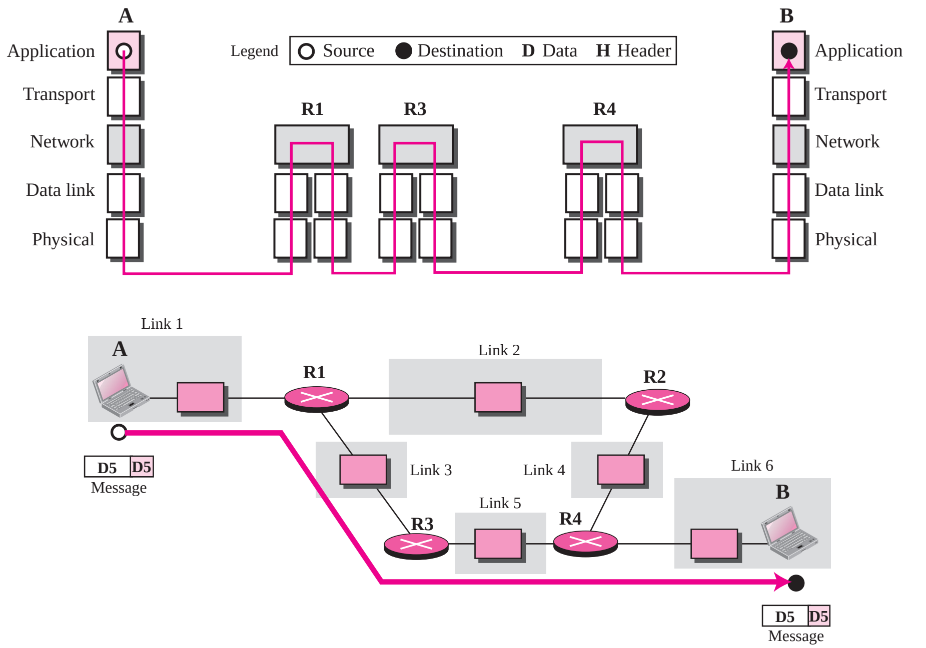
Figure 2.14 *Communication at the application layer*



Note that the communication at the application layer, like the one at the transport layer, is end to end. A message generated at computer A is sent to computer B without being changed during the transmission.

The unit of communication at the application layer is a message.

2.4 ADDRESSING

Four levels of addresses are used in an internet employing the TCP/IP protocols: **physical address, logical address, port address,** and **application-specific address.** Each address is related to a one layer in the TCP/IP architecture, as shown in Figure 2.15.

Physical Addresses

The physical address, also known as the link address, is the address of a node as defined by its LAN or WAN. It is included in the frame used by the data link layer. It is the lowest-level address. The physical addresses have authority over the link (LAN or WAN). The size and format of these addresses vary depending on the network. For example, Ethernet uses a 6-byte (48-bit) physical address that is imprinted on the network interface card (NIC). LocalTalk (Apple), however, has a 1-byte dynamic address that changes each time the station comes up.